# Archaeological Reconstruction of a Metabolic Revolution: Jared P. Rutter and the Discovery of the Mitochondrial Pyruvate Carrier

## A) ANCHOR SCENE CANDIDATES

SCENE #1: The Yeast "Spot Test" and the Resolution of a Forty-Year Biochemical Enigma Summary: The identifying moment of the Mitochondrial Pyruvate Carrier (MPC) involved a fundamental genetic screening process in *Saccharomyces cerevisiae*. For decades, the identity of the protein responsible for transporting pyruvate across the inner mitochondrial membrane remained the "Holy Grail" of mitochondrial biochemistry.[1] In the Rutter laboratory, researchers utilized a systematic approach to functionally annotate uncharacterized mitochondrial proteins.[2] The "Eureka" moment was captured through a simple biological assay: the yeast "spot test." By deleting a conserved but unknown gene (then labeled "Gene X"), the team observed a binary survival outcome.[3] While the mutant yeast thrived on glucose—relying on cytosolic fermentation—they were completely unable to grow on non-fermentable carbon sources that require mitochondrial respiration, such as glycerol or ethanol.[1] This visual evidence of a growth defect provided the first definitive link between these specific proteins and the entry of pyruvate into the "cellular power plant".[5] Date/Place: Early 2012; University of Utah, Department of Biochemistry, Salt Lake City, Utah.[3] People: Jared P. Rutter (Principal Investigator), Daniel K. Bricker (Lead Author), John C. Schell (MD/PhD candidate), and the collective Rutter Lab team.[1] Concrete texture: The workspace was characterized by a high-throughput integration of classical yeast genetics and modern mass spectrometry.[3] The specific "texture" of the discovery was found on synthetic media plates where serial dilutions of yeast cultures were "spotted" in rows.[4] The "wild-type" yeast rows grew into robust, opaque circles, while the *mpc1Δ* and *mpc2Δ* mutant rows remained empty or showed only faint, dying cells.[4] Subsequent analysis involved "crushing" these yeast cells and subjecting the extract to gas chromatography-mass spectrometry (GC-MS) using an Agilent 5977B or a similar high-resolution Orbitrap instrument.[1] The data revealed a massive peak for pyruvate ( $C_3H_4O_3$ ) and a corresponding valley for TCA cycle intermediates like citrate ( $C_6H_8O_7$ ) and $\alpha$-ketoglutarate ( $C_5H_6O_5$ ).[1] Why it matters: This scene represents the culmination of a search that began in the early 1970s.[1] By identifying MPC1 and MPC2, the Rutter lab provided the molecular handle necessary to study metabolic reprogramming in cancer (the Warburg Effect), heart failure, and diabetes.[3] It transformed an abstract biochemical concept into a tangible genetic target for drug development.[2] Evidence quality:

Primary. The results are extensively documented in the 2012 *Science* publication and visualized in supplementary data.[4] Myth risk: Low. The "spot test" is a standard, repeatable laboratory procedure, and the biochemical blockage was quantitatively verified through mass spectrometry.[3] Sources:.[1]

SCENE #2: The Clinical Bridge: Sequencing the Brivet Patient Summary: Following the discovery in yeast and Drosophila, the Rutter lab sought to validate the human relevance of the MPC.[1] They engaged in a retrospective "medical detective" search, identifying a 2003 clinical report by Michèle Brivet and colleagues describing a young girl with an undiagnosed metabolic disorder.[15] The girl suffered from severe lactic acidosis and hyperpyruvatemia, symptoms that mirrored the Rutter lab's MPC-deficient yeast.[16] Rutter and his team contacted the clinicians to obtain DNA samples from the patient's family, who had been waiting for answers for nearly a decade.[1] Date/Place: Circa late 2011; Collaboration between the University of Utah (USA) and researchers in France.[1] People: Jared Rutter, John Schell, Carl Thummel (collaborator in human genetics), and Michèle Brivet (original clinician).[5] Concrete details: The clinical case involved a girl born to consanguineous parents, who exhibited "failure to thrive," hepatomegaly, and severe psychomotor retardation, leading to her death at 19 months.[16] Sequencing the patient's DNA revealed a homozygous 289C-T transition in the MPC1 gene.[16] This mutation resulted in an Arginine-to-Tryptophan substitution ( $R97W$ ) at a residue that is conserved across billions of years of evolution, from yeast to humans.[16] In the lab, Rutter's team demonstrated that expressing the mutant human protein in yeast failed to rescue their growth defect, whereas the wild-type human protein restored growth, definitively linking the patient's death to this single genetic transporter.[1] Why it matters: This scene highlights the "bench-to-bedside" impact of basic science.[7] It provided closure for the family of the patient and established a new class of human metabolic diseases: mitochondrial pyruvate carrier deficiency (MPYCD).[16] Evidence quality: Primary. The genetic data is archived in the ClinVar database (Variation ID 35561) and detailed in the 2012 *Science* paper.[9] Myth risk: Medium. While the sequencing and rescue experiments are documented, the "moment of connection" is often dramatized in talks to emphasize the 10-year gap between the clinical case and the genetic solution.[1] Sources:.[1]

SCENE #3: The Independent Fellow and the Discovery of PAS Kinase (PASK) Summary: Before his work on the MPC, Rutter's career was defined by a period of intense autonomy as the first "Sara and Frank McKnight Independent Fellow" at UT Southwestern.[7] This role allowed him to bypass the traditional postdoctoral apprenticeship and launch his own independent research program immediately after his PhD.[7] He focused on the PAS domain—an ancient sensory module—and identified PAS kinase (PASK) as a critical link between cellular energy sensing and the regulation of glycogen synthesis.[21] Date/Place: 2001–2003; University of Texas Southwestern Medical Center, Dallas, Texas.[7] People: Jared Rutter and Steven L. McKnight (PhD advisor and department chair).[7] Concrete details: Rutter utilized low-stringency BLAST

searches of the GenBank database to identify proteins that shared structural motifs with the *Bradyrhizobium japonicum* FixL PAS domain.[22] The discovery involved characterizing hPASK, a polypeptide containing two N-terminal PAS domains followed by a serine/threonine kinase domain.[22] He used truncation mutants, such as $\Delta 444$ (lacking the PAS domain) and $\Delta 955$ (lacking the mid-region), to demonstrate that the PAS-A domain acts as a *cis*-inhibitor of the kinase activity.[21] This biochemical work was conducted in HEK293 cells using co-immunoprecipitation with anti-V5 antibodies to show the physical interaction between PASK and its substrate, glycogen synthase.[21] Why it matters: This era established Rutter's core scientific philosophy: that metabolism is not just a collection of pathways but a sensory system that makes "decisions".[20] It proved his ability to lead an independent laboratory and set the stage for his recruitment to the University of Utah.[7] Evidence quality: Primary. Documented in *PNAS* (2001, 2005) and confirmed by institutional biographies.[7] Myth risk: Low. The Fellowship is a matter of institutional record, and the BLAST-search methodology is a standard bioinformatic approach.[22] Sources:.[7]

SCENE #4: The Launch of the MIDAS Platform and Systematic Allostery Summary: Seeking to move beyond the study of single proteins, Rutter initiated a project to map the entire landscape of metabolite-protein interactions.[2] This led to the creation of the MIDAS (Mass spectrometry Integrated with equilibrium Dialysis for the discovery of Allostery Systematically) platform.[3] The scene involves the transition of the Rutter lab from a traditional molecular biology group to a technology-driven "omics" powerhouse.[2] Date/Place: Circa 2018–2020; University of Utah, Salt Lake City, Utah.[2] People: Jared Rutter, Kevin Hicks (Research Assistant Professor and key collaborator).[2] Concrete details: The MIDAS system relies on the principle of equilibrium dialysis, where a semi-permeable membrane separates a protein solution from a library of metabolites.[3] The "texture" of the work involves high-resolution mass spectrometry using Orbitrap instruments to identify which small molecules have "stuck" to a particular protein.[10] In their first massive push, the team screened over 200 proteins, uncovering hundreds of previously unknown allosteric interactions.[2] This technological shift culminated in the founding of Atavistik Bio, a company aimed at drugging these newly discovered metabolic nodes.[7] Why it matters: This represented a major evolution in Rutter's career, moving from identifying missing transporters to building a systematic framework for understanding how the "words" of metabolism (metabolites) control the "actors" of the cell (proteins).[2] Evidence quality: Primary. Featured in institutional research statements and recent grant accomplishments.[2] Myth risk: Low. The technology is active and central to his current HHMI research portfolio.[2] Sources:.[2]

# B) DIRECT QUOTES FROM THE SCIENTIST

1. "I have become convinced that the metabolic situation of the cell is a really important environmental factor that, until recently, was largely overlooked as anything other than just providing ATP and enabling the cell to go about and do its job. I have become

convinced that metabolism is much more active in determining what the cell does rather than just doing what it's told." Context/Date: Interview with ASBMB Today, October 1, 2015. Source:.[23]

2. "I think one thing that was clear, and continues to be clear, is that there's a lot that we don't know about how metabolism and metabolites signal to control cellular behaviors and cellular decisions. However, I think there are a lot of unanswered questions in most fields, so I don't know that there was anything particularly unique about it other than that's the lab I ended up in, and I found a lot of exciting things to do and so I was therefore excited to keep doing it." Context/Date: Interview with *Development* journal, October 25, 2023. Source:.[19]
3. "While the pandemic was horrible for many people around the world, one benefit that I experienced was that I started playing more golf, which has become my major pastime outside of the lab now that my kids have left home. I would play more golf if I were less busy with work, but I enjoy the work I do." Context/Date: "Pioneer in Metabolism" interview with UMass Chan Medical School, June 11, 2024. Source:.[11]
4. "In my fantasies, I would have been a professional golfer." Context/Date: Response to "What profession do you think you would have if you weren't a scientist?" in June 2024 interview. Source:.[11]
5. "I would encourage young scientists to be thoughtful and intentional about becoming the scientist they want to be. Build the skill set and tools that you want to have. Do your best to make yourself into the ideal person for the job that you want to have. The most direct path for most people will be to discover something important." Context/Date: Career advice section of the MetNet interview, June 11, 2024. Source:.[11]
6. "Doing what we do is hard but having people who are battling alongside you makes it easier." Context/Date: Describing the collaborative culture at the University of Utah in 2024. Source:.[11]
7. "We have identified the functions of several previously uncharacterized mitochondrial proteins, including the discovery of the long-sought mitochondrial pyruvate carrier... This knowledge has now enabled for the first time the demonstration that this critical metabolic step is impaired in a variety of human diseases." Context/Date: Summary of research interests on the Atavistik Bio and University of Utah faculty sites. Source:.[3]

# C) HUMAN REALITY DETAILS

- **Scientific Heredity:** Rutter is the nephew of William "Bill" Rutter, a titan of 20th-century biochemistry who served as the chair of Biochemistry and Biophysics at UCSF and co-founded the Chiron Corporation, a pioneer in viral diagnostics.[23]
- **Childhood Inspiration:** He credits his early interest in nature to PBS nature shows, specifically the underwater explorations of Jacques Cousteau, which he watched as a young child.[20]
- **The "Independent Fellow" Model:** His career was launched by the unique Sara and Frank McKnight Fellowship, which provided him with 18 months of complete scientific

autonomy and resources before he even joined a faculty, allowing him to bypass the traditional subordinate postdoctoral stage.[7]

- **Athletic Passion:** Rutter is an avid golfer and actively seeks out opportunities to play during scientific conferences; he famously mentioned planning to play in San Diego during an ASBMB meeting.[23]
- **Musical Success in the Family:** His son, Jack Rutter, is the acclaimed indie-pop musician known as "Ritt Momney," whose cover of Corinne Bailey Rae's "Put Your Records On" reached nearly a billion streams and became a global TikTok phenomenon in 2020.[11]
- **Parental Influence on Art:** Jack Rutter (the son) once recalled in an interview that his parents (Jared and his wife) had "pushed" him into a role as a "Mormon movie star" child actor, for which he received seventy-five dollars that was placed in a savings account he didn't know existed.[30]
- **Pandemic Evolution:** During the COVID-19 lockdowns, Jared transitioned his leisure time to focus intensely on golf, as his children had left the family home, leaving him with more free time outside the laboratory.[11]
- **Mormon Cultural Context:** While Jared works at the University of Utah, his son's music often explores the "drift away from Mormon culture" and the personal struggles of growing up in the unique religious landscape of Salt Lake City.[30]
- **Collaborative Philosophy:** He describes the University of Utah's metabolism community as one where people are "genuinely excited for each others' successes," contrasting it with more hyper-competitive academic environments.[11]
- **Alternative Career Path:** If he had not pursued science, Rutter admits he would likely have become an engineer—a choice he jokingly characterizes as "boring" compared to his "fantasy" of being a pro golfer.[11]
- **Mentorship Legacy:** He was mentored by Steven L. McKnight, a scientist-entrepreneur who returned to academia after leading a biotech company, a background that clearly influenced Jared's own comfort with translating academic research into private companies like Atavistik Bio.[7]
- **Lab Instrument Specifics:** His team relies on highly specialized hardware including the Seahorse XFe96 Analyzer for measuring cellular respiration and the Thermo Scientific Q Exactive Plus Orbitrap for precise mass spectrometry.[10]
- **Institutional Loyalty:** Despite being an HHMI Investigator with national prestige, Rutter has remained at the University of Utah for his entire faculty career (since 2003), often praising the local skiing, biking, and hiking culture.[8]

# D) SCIENCE CONNECTION TO INTRO BIOCHEMISTRY (BIOL 3030 BRIDGE)

## Best-Fit Topic: Metabolism and the Central Role of Pyruvate

The work of Jared Rutter is most effectively integrated into introductory biochemistry curricula

during the study of the Citric Acid Cycle (TCA cycle) and mitochondrial membrane transport.

### Precise Contribution

Jared Rutter identified the molecular identity of the Mitochondrial Pyruvate Carrier (MPC), the heterodimeric protein complex (MPC1 and MPC2) that transports pyruvate from the cytosol across the inner mitochondrial membrane into the matrix.[3]

### Teachable Bridge Explanation

In standard biochemistry, students learn that glycolysis occurs in the cytoplasm, resulting in the production of pyruvate:

$$\text{Glucose} + 2\text{NAD}^+ + 2\text{ADP} + 2\text{P}_i \rightarrow 2\text{Pyruvate} + 2\text{NADH} + 2\text{H}^+ + 2\text{ATP}$$

However, the subsequent conversion of pyruvate into Acetyl-CoA by the Pyruvate Dehydrogenase (PDH) complex occurs strictly within the mitochondrial matrix.[1] Because the inner mitochondrial membrane is impermeable to polar molecules, a specific transporter is required to move pyruvate into the mitochondria.[1] Rutter discovered that the MPC1/MPC2 complex acts as this "gatekeeper"; without it, pyruvate is diverted away from the TCA cycle and converted into lactate, a phenomenon frequently observed in the "Warburg Effect" of cancer cells.[1]

### Scientist's Own Framing/Analogy

Rutter and his lab members frequently use the analogy of the "cell's engine" to describe mitochondria, framing the MPC as the "fuel intake valve" that determines whether the cell will "burn" carbohydrate for energy or convert it to waste products.[1] He often describes mitochondria as "the mysterious cellular parasite," referencing their endosymbiotic origins and their autonomous role in signaling.[6]

## E) COMPLICATIONS, DISPUTES, OR CONTROVERSIES

### Item: Simultaneous Discovery and Priority Coordination (2012)

**What's agreed:** In May 2012, two independent research groups published papers in the same issue of the journal *Science* identifying MPC1 and MPC2 as the components of the mitochondrial pyruvate carrier.[9] The first was the Rutter/Thummel team (Bricker et al., 2012), and the second was a European collaboration involving Jean-Claude Martinou and Stefan Bröer (Herzig et al., 2012).[9] **What's disputed:** While the scientific community generally acknowledges both groups as co-discoverers, institutional narratives often emphasize different "founding moments".[1] The Rutter lab's narrative focuses heavily on the clinical connection to the Brivet patient, which provided a human genetic "proof" that was distinct

from the yeast-focused discoveries of the other team.[1] There is no evidence of a hostile priority dispute, but the simultaneous publication indicates a fierce international race for the molecular identity of this decades-old carrier.[32] **Sources:**.[1]

### Item: The Confusion of the "John Rutter" Name

**What's agreed:** Jared Rutter's son is a famous musician.[11] **What's disputed:** A widespread secondary-source confusion exists regarding the musician's name. John Rutter is a world-renowned British choral composer (born 1945), whereas Jared Rutter's son is Jack Rutter (born circa 2000), who performs under the stage name Ritt Momney.[28] Many casual observers and some journalistic snippets conflate the two due to the shared surname and the father's mentions of a "musician son".[11] **Sources:**.[28]

### Item: The Effectiveness of the "Independent Fellowship" Model

**What's agreed:** Jared Rutter was the inaugural recipient of the McKnight Independent Fellowship at UT Southwestern.[7] **What's disputed:** While this model was a massive success in Rutter's case, the broader academic community continues to debate whether bypassing the traditional postdoctoral mentor-mentee relationship is sustainable or beneficial for most young scientists.[19] Rutter himself admits that this path required a "lot of freedom and flexibility" that was only possible because his PhD mentor, Steve McKnight, had significant resources and no "territorial" claims on the projects Jared was pursuing.[20] **Sources:**.[7]

## F) "NOT ON WIKIPEDIA" GOLD

1. **Finding:** Rutter's conference presentation title "Mitochondria: The Mysterious Cellular Parasite".[6] **Why it is distinctive:** This title reveals a specific intellectual framing of mitochondria as autonomous biological actors rather than mere passive organelles, a theme that pervades his research on "metabolic decisions".[6] **Source:**.[6]
2. **Finding:** The detail that his son Jack (Ritt Momney) recorded the viral billion-stream hit "Put Your Records On" in the basement of Jared's Salt Lake City home during the COVID-19 quarantine.[29] **Why it is distinctive:** It provides a domestic, human-scale backdrop to the home life of a world-leading scientist, showing the intersection of high-level biochemistry and modern internet pop culture.[29] **Source:**.[29]
3. **Finding:** Rutter was the *first* scientist to demonstrate that PAS domains—previously known mostly as sensors of light and oxygen in bacteria—are also used in human cells to sense metabolic status through the PASK enzyme.[21] **Why it is distinctive:** This discovery proved that metabolic sensing is an evolutionarily ancient process that has been conserved across billions of years of life, from soil bacteria to human muscle.[22] **Source:**.[22]
4. **Finding:** The specific biochemical hardware listed in his laboratory's training grants: "Agilent 5977B GC/MSD" and "Thermo Scientific Q Exactive Plus Orbitrap Mass Spectrometer".[10] **Why it is distinctive:** These technical details provide the "concrete

texture" of how metabolic discoveries are actually measured, moving the narrative from abstract "discovery" to the reality of high-precision instrumentation.[10] **Source:**.[10]

5. **Finding:** Rutter's role as the first-ever "Sara and Frank McKnight Independent Fellow," a role that essentially created a new career pathway in American biological sciences.[7] **Why it is distinctive:** Most prominent scientists have a "famous postdoc mentor"; Rutter is unique in that his most famous work was done in his own name as a 28-year-old independent researcher.[7] **Source:**.[7]

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# H) FLAGGED “MYTH-RISK” DETAILS

**Claim:** "The discovery of the MPC was a 10-year journey to solve the mystery of the little girl who died." **Where it appears:** iBiology presentation by John Schell and institutional news releases.[1] **Why sourcing is weak:** While the Brivet clinical paper was published in 2003 and the MPC discovery in 2012 (a 9-year gap), the narrative often frames it as a singular, decade-long quest specifically to solve *that* patient's case.[1] In reality, the Rutter lab was systematically screening uncharacterized mitochondrial proteins; the discovery came first, and the connection to the Brivet paper was made *retrospectively* to provide human validation.[3] The "10-year search" is a narrative rounding that emphasizes drama over the serendipity of the literature search.[16] **Best available evidence:** The 2012 *Science* paper confirms the sequencing of the families happened at the end of the project to confirm the biological results found in yeast and flies.[9]

**Claim:** "Jared Rutter is an expert professional golfer." **Where it appears:** Secondary mentions in blog comments and career fantasy questions.[11] **Why sourcing is weak:** Rutter himself explicitly labels this a "fantasy".[11] He is a highly skilled and enthusiastic amateur, but the phrasing "would have been a professional golfer" is a response to a hypothetical question about alternative lives.[11] There is no record of professional play or ranking. **Best available evidence:** ASBMB profile confirms he is an "avid golfer" who plays during meetings.[23]

**Claim:** "Jared Rutter is the brother/son of John Rutter, the composer." **Where it appears:** General internet search confusion and social media speculation.[28] **Why sourcing is weak:** This is a case of homonym confusion. John Rutter (composer) is British; Jared Rutter is American.[34] There is no documented biological relationship between the composer John Rutter and the scientist Jared Rutter.[28] **Best available evidence:** Family details confirm Jared is the nephew of Bill Rutter (scientist/entrepreneur) and the father of Jack Rutter (indie musician).[26]

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# LENS RECOMMENDATION

## Lens Scoring Rubric

| Lens | Score | Justification |
|---|---|---|
| **Eureka (Discovery)** | 3 | The yeast "spot test" provides a perfect, binary |

| | | |
|---|---|---|
| | | visual of discovery (growth vs. no-growth) that is highly effective for students.[3] |
| **Failure (Setback)** | 1 | Rutter's career is remarkably linear and successful; no major setback "changed his trajectory" as much as his early success.[7] |
| **Mentor (Relationship)** | 2 | The relationship with Steve McKnight is significant, but it was a relationship of "independence" rather than traditional shaping.[19] |
| **Controversy (Conflict)** | 2 | The simultaneous discovery with the Bröer lab provides a "race" narrative, but lacks the personal drama of a "dispute".[32] |
| **Human Detail** | 3 | The golfer persona paired with the "billion-stream" viral musician son provides an extraordinary human-interest angle.[11] |
| **Last Experiment** | 0 | Scientist is still active and mid-career.[7] |

## Primary Recommended Lens: Eureka (Discovery Crystallized)

**Why it fits:** This lens is the best choice for an introductory biochemistry audience (BIOL 3030) because it ties a complex biochemical problem—pyruvate transport—to a simple, visual experimental result (the yeast spot test) and a high-stakes clinical payoff (the Brivet patient). It allows the teacher to explain the fundamental logic of "phenotype vs. genotype" while highlighting the importance of basic research in solving human disease.[1]

- **Best opening scene:** Scene #1 (The Yeast Spot Test).[4]
- **One quote to use early:** Quote #1 ("Metabolism is... more active in determining what the cell does rather than just doing what it's told").[23]

- **Cautions:** Avoid claiming Rutter set out in 2002 specifically to find the gene for the "little girl"; emphasize that the clinical link was the "Aha!" moment that happened after the yeast discovery.[9]

## Backup Lens: Human Detail

**Why it fits:** If the class profile needs to be more "humanistic" or approachable, focusing on the "Family of Creators" angle is very strong. Students will likely recognize the music of Jack Rutter (Ritt Momney). Framing Jared as a scientist who balances high-stakes biochemical research with a love of golf and a family life that produced a pop-culture phenomenon is a powerful way to break the "sterile scientist" stereotype.[11]

- **Best opening scene:** Scene #4 (The basement studio vs. the laboratory).[29]
- **One quote to use early:** Quote #4 ("In my fantasies, I would have been a professional golfer").[11]
- **Cautions:** Ensure the distinction between Jack Rutter (son) and John Rutter (unrelated composer) is clear to prevent spreading the common name-myth.[28]

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